

TELCO DIGITAL CREDIT PLATFORM

Software Requirements Specification (SRS) - Version 2.0

Multi-Telco Airtime, Data and Digital Credit Advance Platform

Document Attribute	Value
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Core principle	Configuration-first, multi-telco, ledger-led, explainable and resilient
Intended use	Coding-agent build specification, vendor RFP baseline, telco integration design and implementation governance

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Document Control

Version	Date	Author/Owner	Summary
1.0	July 2026	Project Team	Initial end-to-end SRS
2.0	16 July 2026	Project Team	Multi-telco architecture, system-of-record boundaries, configuration-first admin, advanced scoring, loan ledger, edge cases, scalability, reconciliation and operational controls

Approval Role	Name	Decision	Date
Executive Sponsor		Pending	
Chief Product Officer		Pending	
Chief Technology Officer		Pending	
Chief Risk Officer		Pending	
Chief Information Security Officer		Pending	
Finance Director		Pending	
Compliance/Data Protection Officer		Pending	
Lead Telco Partner		Pending	

1. Executive Summary

This SRS defines a production-grade digital credit platform that can support multiple telecommunications operators, beginning with airtime and data advances and extending to other operator-distributed credit products. The platform shall ingest behavioural and subscriber-status data from each telco, calculate eligibility and limits at scale, return offers with low latency, instruct or coordinate fulfilment through telco systems, maintain authoritative advance and repayment records, reconcile all activity, and calculate settlements and revenue shares.

The platform shall be operator-neutral. MTN, Airtel, Globacom, 9mobile, MVNOs and future operators shall connect through independently configurable adapters behind a Telco Integration Gateway. Every tenant-owned entity shall carry an immutable telco identifier. A telco outage, data defect or settlement dispute shall be isolated from other operators.

The design is configuration-first. Product rules, tiers, score weights, fee rates, revenue shares, operational thresholds, reconciliation tolerances, workflow rules, feature flags and most integration behaviour shall be manageable through an authorised administration portal with versioning, approval, testing and rollback. Core accounting invariants, security controls and data-integrity rules shall not be bypassable through configuration.

2. Purpose and Objectives

The purpose is to provide a single scalable platform for telco-assisted digital credit while preserving clear ownership boundaries between telco systems, the lending platform and finance systems.

Primary objectives are instant customer decisions, controlled credit risk, accurate recovery, complete ledgering, reliable reconciliation, configurable multi-telco onboarding, auditable decisions, safe product experimentation and operational resilience at tens of millions of subscribers and high transaction volumes.

3. Scope

3.1 In Scope

- Multi-telco onboarding and tenant management.
- Subscriber data ingestion through batch files, streaming events and synchronous APIs.
- Eligibility, scoring, affordability, trust progression, anti-gaming and limit assignment.
- Airtime, data, voice bundle and configurable digital advance products.
- Offer retrieval and low-latency decisioning.
- Advance creation, fulfilment orchestration, exposure management and immutable ledgering.
- Recharge-triggered and other configured recovery mechanisms.
- Repayments, partial repayments, reversals, adjustments, refunds, write-offs and recoveries.
- Multi-party fees, commissions, tax, funding cost and settlement calculations.
- Reconciliation, exception management, disputes and operational case management.
- Customer, telco, lender/funder, operations, risk, finance, support and administration interfaces.
- Security, privacy, audit, observability, disaster recovery and regulatory controls.

3.2 Out of Scope for Initial Release

- Direct control of a telco subscriber airtime or data balance except through authorised telco APIs.
- Replacement of telco CRM, billing, charging, network or identity systems.
- General-purpose cash lending unless separately approved as a product.
- Credit bureau reporting unless enabled through a later approved integration.
- Manual editing of immutable ledger entries.

4. Architectural and Product Principles

Principle	Required Interpretation
Multi-telco by design	No core service shall contain MTN-specific logic. Operator differences shall be isolated in adapters and configuration.
Configuration-first	Business policies shall be administered without code deployment wherever safe.
Ledger-led	Financial truth shall be derived from append-only ledger entries, not mutable balances alone.
Telco-assisted, platform-owned credit	Telcos own network-side balances and source events; the lending platform owns credit decisions, advances, exposure and recovery allocation.
Idempotent and replay-safe	Duplicate requests and event replays shall not create duplicate loans, credits or repayments.
Explainable decisions	Every approval, decline and limit shall be reproducible from stored inputs, rule versions and model versions.
Fail safely	Uncertain fulfilment or recovery states shall be quarantined and reconciled, not guessed.
Tenant isolation	A failure, attack or misconfiguration for one telco shall not compromise another.
Progressive trust	Limit growth shall be earned over time; one scoring cycle may increase a customer by no more than the configured tier movement, default one tier.
Privacy minimisation	Use flags and derived attributes rather than raw identity data where sufficient.

5. Systems of Record and Ownership Boundaries

System	Authoritative For	Not Authoritative For
Telco System of Record	MSISDN status, SIM registration/NIN-verification flags, subscriber activity, source usage/recharge data, airtime/data balances, fulfilment and network-side deduction events	Credit score, advance contract, exposure, repayment allocation, platform ledger
Lending Platform System of Record	Eligibility, score, limit, offers, advance records, exposure, repayment allocation, ledger, reversals, write-offs, reconciliation and commercial settlement calculations	Telco subscriber balance or statutory GL
Finance System of Record	General ledger, cash settlement, revenue recognition, statutory reporting, audited financial statements	Real-time offer and telco fulfilment state

The platform shall retain complete advance records even when the telco performs the actual airtime/data credit. The telco fulfilment reference and platform advance identifier shall be cross-linked and reconciled.

6. Stakeholders and User Roles

Role	Responsibilities
Platform Super Administrator	Cross-telco configuration, tenant onboarding, emergency controls and platform-wide oversight.
Telco Administrator	Operator-specific products, users, reports and integration settings within delegated permissions.
Risk Analyst/Manager	Rules, scoring weights, tier policies, exposure caps, monitoring and model governance.
Finance/Settlement User	Fees, commissions, reconciliation, settlement statements and finance exports.
Operations User	Exceptions, fulfilment failures, recovery issues, batch processing and cases.
Customer Support Agent	Subscriber history, explanations, complaints and controlled adjustments.
Fraud Analyst	Fraud alerts, SIM-swap events, linked identities and account restrictions.
Compliance/Audit User	Read-only access to decisions, versions, consents, audit events and reports.
Funding Partner User	Portfolio and settlement views limited to funded products and telcos.
Subscriber	Requests offers/advances, receives disclosures, views status and receives notifications.

7. Multi-Telco Architecture

7.1 Tenant Identification

Every subscriber account, score, offer, advance, repayment, ledger entry, reconciliation item, settlement item, case, audit event and configuration object shall carry an immutable telco_id. MSISDN alone shall never be the primary business key. The canonical subscriber key shall be telco_id + normalised MSISDN + effective service period, with internal subscriber_account_id.

7.2 Telco Integration Gateway

- Authenticates each telco using operator-specific credentials and certificates.
- Resolves telco context from credentials and payload; rejects inconsistent identifiers.
- Transforms operator-specific schemas and codes into canonical platform messages.
- Applies per-telco rate limits, timeouts, retries, circuit breakers and maintenance modes.
- Routes outbound fulfilment, enquiry and acknowledgement messages to the correct adapter.
- Maintains independent dead-letter queues and replay controls for each operator.

7.3 Messaging and Routing

The platform shall support shared logical topics partitioned by telco_id and dedicated operator topics where volume, security or isolation requires it. The routing strategy shall be configurable by operator and event type. Partition keys shall preserve ordering where needed, especially for subscriber-level repayment and fulfilment events.

ID	Requirement	Priority	Acceptance Evidence	Configurable?
MT-001	All inbound and outbound requests shall include or resolve an immutable telco_id.	Must	Cross-tenant request tests pass.	No
MT-002	Each telco shall have independently configurable endpoints, credentials, queue/topic names, limits and failure policies.	Must	Onboard second telco without core-code change.	Yes
MT-003	A telco adapter failure shall not consume all shared worker, connection or queue capacity.	Must	Chaos test demonstrates continued service for other telcos.	Partly
MT-004	Administrative users shall be restricted to permitted telco tenants.	Must	RBAC and row-level security tests.	Yes
MT-005	The platform shall support shared, schema-isolated or dedicated database deployment modes by telco.	Should	Deployment runbook and migration test.	Yes

8. Configuration and Rules Management Framework

The platform shall implement governed configuration, not uncontrolled editable values. Every material configuration shall have scope, effective dates, version, owner, approver, status, validation, impact preview, audit history and rollback capability.

Configuration Domain	Examples
Tenant	Telco code, country, currency, time zone, operational status, data residency.
Integration	URLs, protocol, auth method, certificates, schemas, timeouts, retries, rate limits, queue names, file locations.
Products	Airtime/data/voice/device product, denominations, fee display, repayment method, eligibility.
Scoring	Feature definitions, weights, transformations, thresholds, model version, override rules.
Tiers and limits	Tier values, max movement, exposure caps, cooling periods, offer denominations.
Anti-gaming	Spike thresholds, winsorisation caps, stability windows, suspicious-pattern rules.
Recovery	Allocation order, partial recovery, protected recharge amount, grace logic, event sources.
Commercials	Fee rate, revenue split, tax, funder return, settlement account and cycle.
Reconciliation	Matching keys, tolerance, ageing, auto-resolution and escalation rules.
Workflow	Maker-checker, approval levels, manual review triggers, SLA and escalation.
Feature flags	Enable product/telco/channel/segment/pilot, kill switches and rollout percentages.
Notifications	Templates, language, channels, triggers and quiet hours.
Retention	Data retention and archival by data class and telco.
Reporting	Scheduled reports, recipients, fields, filters and formats.

ID	Requirement	Priority	Acceptance Evidence	Configurable?
CFG-001	Material configuration changes shall require maker-checker approval.	Must	Approval workflow and audit test.	No
CFG-002	Configuration shall support draft, simulation, approved, scheduled, active, superseded and rolled-back states.	Must	Lifecycle tests.	No
CFG-003	The platform shall prevent activation of invalid or internally inconsistent configuration.	Must	Negative validation tests.	No
CFG-004	Each decision and transaction shall retain the exact configuration version used.	Must	Historical replay reproduces result.	No
CFG-005	Emergency kill switches shall be available per telco, product and channel.	Must	Operational drill.	Yes
CFG-006	Secrets shall not be displayed in plain text in the admin portal.	Must	Security test.	No

9. Telco Data Ingestion and Data Contracts

9.1 Supported Interfaces

- Nightly or intraday batch files for large subscriber feature sets.
- Streaming events for recharge, repayment deduction, SIM swap, fraud, barring, NIN-verification changes and fulfilment status.
- Synchronous APIs for eligibility checks, offer retrieval, fulfilment and status enquiries.
- Secure object storage, SFTP, private API connectivity, VPN or dedicated links as agreed.

9.2 Minimum Canonical Subscriber Attributes

Category	Illustrative Attributes
Identity/status	telco_id, normalised MSISDN, subscriber status, prepaid/postpaid, SIM age, registration status, NIN verified flag, KYC tier.
Recharge	Counts, amounts, medians, percentiles, recency, channel, reversals and rolling windows.
Usage	Voice, data, SMS, VAS usage and active days in rolling windows.
Risk events	SIM swap, porting, barring, fraud flag, abnormal recharge, device change and account linkage.
Credit history	Prior advances, recovery amounts, repayment speed, delinquency, write-off and dispute outcomes.
Data quality	Source timestamp, extract timestamp, schema version, completeness and confidence indicators.

ID	Requirement	Priority	Acceptance Evidence	Configurable?
DATA-001	Actual NIN values shall not be required where a verified flag is sufficient.	Must	Data contract review.	Yes
DATA-002	Every feed shall include source time, extract time, telco_id, schema version and unique file/event identifier.	Must	Contract validation.	No
DATA-003	Batch files shall be checksum-validated, decryptable, schema-validated and reconciled to control totals before use.	Must	Failed-file test.	No
DATA-004	Late, duplicate and out-of-order events shall be handled idempotently.	Must	Replay test.	No
DATA-005	Data-quality failures shall not silently overwrite a previously valid subscriber profile.	Must	Quarantine and fallback test.	Partly

10. Eligibility, Scoring, Affordability and Limit Assignment

10.1 Processing Model

The default architecture shall use pre-calculated subscriber eligibility and limits on a scheduled basis, combined with real-time overlays for critical changes. The real-time request path shall normally read an approved offer snapshot rather than recompute all historical features. Full or partial recalculation may be triggered by configured events.

- Scheduled portfolio scoring: daily by default, configurable by telco and segment.
- Event-driven overlays: repayment completion, SIM swap, NIN status change, barring, fraud alert, material recharge, tenure threshold and manual restriction.
- On-demand re-score: authorised operations/risk action with full audit.
- Stale-score policy: configurable maximum score age and safe fallback behaviour.

10.2 Scoring Dimensions

Dimension	Examples	Role
Eligibility gates	Prepaid status, SIM age, active status, NIN verification, fraud/barring, minimum history.	Hard pass/fail before scoring.
Behaviour score	Recharge consistency, usage stability, active days, tenure and recency.	Predictive behavioural quality.
Trust score	Successful advance cycles, recovery speed, partial repayments and defaults.	Progressive limit growth.
Affordability	Stable recharge capacity, expected recovery and exposure-to-spend ratio.	Caps maximum sustainable exposure.
Fraud score	Spikes, cycling, linked accounts, SIM changes, device patterns and suspicious channels.	Decline, reduce or review.
Portfolio controls	Telco/product/funder exposure, vintage performance and concentration.	Portfolio-level cap.

10.3 Anti-Gaming Requirements

- A single large recharge shall not immediately increase a limit materially.
- Features shall use rolling windows, median/trimmed mean, percentiles, variance and stability measures.
- Recharge spikes shall be compared with the subscriber baseline, peer segment and channel behaviour.
- Configured outliers may be capped, discounted, quarantined or excluded from affordability.
- Circular or refunded recharges, agent self-funding patterns and rapid recharge-borrow sequences shall be detected.
- Tier increases shall be limited to the configured maximum per scoring cycle; default one tier.
- Tier decreases may occur immediately where risk increases, subject to customer-treatment rules.
- New subscribers shall begin at conservative starter tiers and earn trust through successful cycles.

ID	Requirement	Priority	Acceptance Evidence	Configurable?
SCR-001	A one-off recharge spike shall not by itself move a subscriber more than the configured maximum tier movement.	Must	Synthetic spike scenario.	Yes
SCR-002	Default maximum upward tier movement shall be one tier per scoring cycle.	Must	Tier transition tests.	Yes
SCR-003	Scoring shall store input snapshot, feature values, score contributions, rules, model and configuration versions.	Must	Decision replay.	No
SCR-004	Scoring shall support champion/challenger	Should	Model governance test.	Yes

ID	Requirement	Priority	Acceptance Evidence	Configurable?
	models and controlled A/B allocation.			
SCR-005	Limits shall be the minimum of risk limit, affordability limit, product cap, telco cap, funder cap and portfolio cap.	Must	Boundary tests.	Partly
SCR-006	The platform shall support reason codes understandable to operations and customers.	Must	Reason-code review.	Yes

11. Product and Offer Management

- Products shall be data-driven and scoped by telco, country, segment, channel and effective date.
- A product may represent airtime, data, voice, device or another approved digital credit item.
- Offer denominations, fee treatment, net value, repayment value, expiry and disclosure shall be configurable.
- Offers shall be generated only from currently active products and the subscriber current available exposure.
- The platform shall support a catalogue of permitted denominations rather than arbitrary values where required by the telco.
- Product changes shall be versioned; existing advances retain the terms accepted at origination.

ID	Requirement	Priority	Acceptance Evidence	Configurable?
PRD-001	Admin users shall create and version products without code deployment.	Must	Create pilot product in test.	Yes
PRD-002	Offer responses shall show gross advance, service fee, net delivered value, total repayment and expiry.	Must	API/UI acceptance test.	Yes
PRD-003	An expired or superseded offer shall not be accepted.	Must	Expiry race test.	No
PRD-004	The platform shall prevent a product from being activated without required settlement and fulfilment configuration.	Must	Validation test.	No

12. Advance Lifecycle and Fulfilment Orchestration

12.1 State Model

State	Meaning
OFFERED	Offer is available but not accepted.
REQUESTED	Customer request received and authenticated.
RESERVED	Exposure reserved; duplicate/concurrent requests blocked.
PENDING_FULFILMENT	Fulfilment instruction submitted or awaiting submission.
ACTIVE	Telco confirmed credit and advance is outstanding.
PARTIALLY_REPAID	Some recovery received.
REPAID	Principal/repayment obligation fully satisfied.
FULFILMENT_FAILED	Telco confirmed no credit; reservation released.
FULFILMENT_UNKNOWN	Outcome uncertain; no retry that could duplicate credit without status enquiry.
REVERSED	Original credit and advance reversed under controlled process.
WRITTEN_OFF	Accounting/risk write-off recorded; later recoveries still supported.
CANCELLED	Request cancelled before fulfilment.
DISPUTED	Operational or customer dispute attached; financial state remains explicit.

12.2 Origination Flow

1. Receive telco/channel request with telco_id, MSISDN, product/amount, request ID and timestamp.
2. Authenticate telco/channel and validate payload.
3. Resolve subscriber account and current offer snapshot.
4. Apply real-time overlay, exposure and product validations.
5. Acquire subscriber-level concurrency control and idempotency record.
6. Reserve exposure and create pending advance record.
7. Post provisional ledger/control entries where applicable.
8. Submit fulfilment to telco or return approval for telco-managed fulfilment according to integration mode.
9. Process definitive telco acknowledgement.
10. Activate or fail the advance and post final ledger entries.
11. Send response/notification and emit downstream events.

ID	Requirement	Priority	Acceptance Evidence	Configurable?
ADV-001	The platform shall maintain an advance record for every approved fulfilment attempt.	Must	Traceability test.	No
ADV-002	Duplicate requests with the same idempotency key shall return the original outcome.	Must	Duplicate storm test.	No
ADV-003	Concurrent requests shall not exceed the subscriber available limit.	Must	Concurrency test.	No
ADV-004	Unknown fulfilment outcomes shall use status enquiry/reconciliation before any repeat credit instruction.	Must	Timeout-after-success simulation.	No
ADV-005	Exposure reservations shall expire or be repaired according to configured policies without losing audit history.	Must	Stuck-reservation test.	Yes

13. Loan Records, Ledger and Exposure

The lending platform shall be the authoritative system for advance records and exposure. It shall maintain both lifecycle records and an append-only financial/sub-ledger. Mutable balance fields may be cached for performance but must be reproducible from ledger entries.

Ledger Event	Illustrative Accounting Meaning
ADVANCE_ORIGINATED	Principal/exposure recognised after confirmed fulfilment.
SERVICE_FEE_RECOGNISED/DEFERRED	Fee treatment based on policy.
REPAYMENT_RECEIVED	Recovery applied to an advance.
REPAYMENT_REVERSAL	Previously reported recovery reversed.
FULFILMENT_REVERSAL	Telco reverses credited value.
ADJUSTMENT	Authorised non-standard correction with reason and approval.
WRITE_OFF	Exposure moved to written-off status.
POST_WRITE_OFF_RECOVERY	Recovery received after write-off.
COMMISSION_ACCRUAL	Telco/funder/platform commercial share calculated.
SETTLEMENT	Obligation settled or netted.

ID	Requirement	Priority	Acceptance Evidence	Configurable?
LED-001	Ledger entries shall be append-only and balanced according to configured accounting event templates.	Must	Ledger integrity tests.	Partly
LED-002	No user shall edit or delete posted ledger entries. Corrections shall use reversal and replacement entries.	Must	Permission and audit test.	No
LED-003	Every entry shall contain telco_id, advance_id, event_id, currency, amount, event time, posting time and source reference.	Must	Schema validation.	No
LED-004	Exposure shall be available by subscriber, telco, product, funder, vintage and portfolio.	Must	Reporting test.	No
LED-005	The platform shall prevent negative exposure or over-repayment except through explicit adjustment workflows.	Must	Boundary tests.	No

14. Recharge Recovery and Repayment Allocation

The telco may intercept or deduct value from recharges and report the recovery, or the platform may calculate a requested deduction that the telco executes. The integration mode shall be configured per telco/product. In both modes, the platform must receive definitive recovery events and maintain the authoritative allocation to advances.

- Support full, partial and multi-event repayment.
- Support configurable allocation order: oldest first, earliest due, product priority, pro-rata or other approved method.
- Support protected recharge portions or minimum retained balances where required.
- Support repayment events from voucher, bank top-up, transfer or other channels as configured.
- Support recovery reversals caused by top-up reversal, chargeback or telco correction.
- Prevent double allocation of replayed events.
- Reopen or adjust advances after a repayment reversal.
- Apply repayment first to components according to configured waterfall, e.g., principal, fee, tax, penalty where legally permitted.

ID	Requirement	Priority	Acceptance Evidence	Configurable?
REP-001	Every recovery event shall be idempotently matched and allocated once.	Must	Replay test.	No
REP-002	Partial recovery shall update outstanding exposure immediately.	Must	Partial repayment test.	No
REP-003	A reversal shall restore the appropriate outstanding amount and retain full history.	Must	Reversal test.	No
REP-004	Unmatched recovery events shall enter an exception queue and never be discarded.	Must	Unknown subscriber/advance test.	No
REP-005	Allocation rules shall be configurable by telco/product with version control.	Must	Policy switch test.	Yes

15. Reconciliation, Settlement and Commercial Accounting

15.1 Reconciliation Layers

Layer	Comparison
Request/decision	Telco requests versus platform responses.
Fulfilment	Platform-approved advances versus telco credits.
Recovery	Telco deductions/recharge recoveries versus platform repayments.
Ledger	Operational records versus sub-ledger totals.
Commercial	Fees and revenue shares versus contract terms.
Settlement	Calculated obligations versus statements and cash received/paid.
Finance	Platform settlement exports versus finance system postings.

15.2 Settlement

- Separate settlement by telco, product, funder, currency and contractual period.
- Support gross, net and hybrid settlement models.
- Support service fee, telco share, platform share, funder return, tax and other deductions.
- Generate statements, supporting transaction files and signed control totals.
- Track payable/receivable ageing, disputes and settlement adjustments.
- Prevent configuration changes from retroactively changing closed settlement periods unless a controlled restatement is approved.

ID	Requirement	Priority	Acceptance Evidence	Configurable?
REC-001	Matching shall use stable external references plus controlled secondary matching rules.	Must	Reconciliation test pack.	Yes
REC-002	Exceptions shall have type, value, owner, age, SLA, status, evidence and resolution.	Must	Case workflow test.	Yes
REC-003	Tolerance-based auto-resolution shall be configurable but fully audited.	Should	Tolerance scenario.	Yes
SET-001	Settlement calculations shall be reproducible from transaction-level data and configuration versions.	Must	Recalculation test.	No
SET-002	Closed periods shall be locked; later changes shall post as adjustments in an open period.	Must	Period-lock test.	Yes

16. Front-End Portals and User Experience

16.1 Administration Portal

- Telco onboarding wizard and tenant status.
- Product, tier, fee, scoring, anti-gaming, recovery, settlement and notification configuration.
- Integration endpoint, schema, certificate metadata, queue/topic and health configuration.
- Feature flags, pilot cohorts, rollout percentages and emergency suspension.
- Maker-checker approvals, effective dating, simulation, impact preview and rollback.
- User, role, telco scope and privileged-access management.
- Audit and configuration-difference views.

16.2 Operations Portal

- Cross-telco dashboard with authorised filters and tenant-specific views.
- Advance and repayment search; subscriber timeline.
- Fulfilment-unknown, stuck reservations, unmatched repayments and batch failures.
- Retry/status-enquiry controls that are safe and idempotent.
- Case assignment, notes, attachments, SLA and escalation.
- Operational health, queue lag and telco connectivity.

16.3 Risk Portal

- Portfolio exposure, approval, take-up, recovery, delinquency and vintage views.
- Tier migration, score distribution, drift and anti-gaming alerts.
- Rule/model simulation before activation.
- Champion/challenger experiment results.
- Overrides, blacklists, whitelists and exposure caps with approvals.

16.4 Finance and Settlement Portal

- Transaction, fee, commission and settlement dashboards.
- Reconciliation exceptions and statements.
- Period close, lock, adjustment and export controls.
- Telco/funder receivable and payable ageing.

16.5 Customer Support and Customer Channels

- Support agents see only necessary masked identity and full transaction history.
- Customer-facing channels may be USSD, SMS, telco app, web, IVR or API.
- Disclosures shall show advance value, fee, net benefit, repayment obligation and recovery mechanism before acceptance.
- Customers shall receive confirmations, repayment notifications, outstanding balance information and complaint channels.

17. API and Integration Requirements

API/Event	Purpose
Get Offers	Return current eligible products and denominations.
Create Advance	Accept selected offer and initiate/reserve fulfilment.
Fulfilment Callback	Receive definitive telco outcome.
Advance Status	Resolve timeouts and unknown outcomes.
Recharge/Recovery Event	Report deduction/recovery.
Recovery Reversal	Reverse previously reported recovery.
Subscriber Status Event	Update barring, NIN, SIM swap, porting or fraud status.
Batch Feature Feed	Provide large-scale scoring inputs.
Reconciliation File	Exchange daily control totals and transaction records.
Settlement Statement	Exchange calculated and acknowledged obligations.

ID	Requirement	Priority	Acceptance Evidence	Configurable?
API-001	All mutation APIs shall require idempotency keys and signed/authenticated requests.	Must	Security and duplicate tests.	No
API-002	Canonical APIs shall be versioned and backward-compatible for an agreed support period.	Must	Contract tests.	Partly
API-003	Responses shall include stable error codes, correlation ID and retry guidance.	Must	Negative API tests.	Partly
API-004	Sensitive fields shall be encrypted in transit and masked in logs.	Must	Security test.	No
API-005	Per-telco rate limiting and quotas shall be configurable.	Must	Load/rate test.	Yes

18. Scalability, Performance and Availability

Area	Requirement
Subscriber scale	Support at least 100 million subscriber profiles across telcos, with horizontal growth.
Scoring throughput	Score tens of millions of subscribers within the configured batch window using partitioned/distributed processing.
Offer latency	Target p95 <= 150 ms and p99 <= 300 ms within the platform boundary for cached offer retrieval, excluding telco network latency.
Advance decision latency	Target p95 <= 300 ms before external fulfilment call for normal path.
Event throughput	Scale horizontally for burst recharge and repayment events; exact capacity shall be established through telco forecasts.
Availability	Core real-time decision services target 99.99% availability; batch/reporting services may have lower defined SLOs.
RPO/RTO	Ledger/advance RPO near zero through synchronous replication or durable log; target RTO <= 30 minutes for critical services, subject to deployment design.
Isolation	Independent autoscaling, quotas and circuit breakers per telco adapter.
Backpressure	Queues and APIs shall shed or defer non-critical work before compromising financial correctness.
Data partitioning	Partition high-volume data by telco_id, time and/or subscriber hash to avoid hotspots.

ID	Requirement	Priority	Acceptance Evidence	Configurable?
NFR-001	No single application node shall be required for correctness or availability.	Must	Failure test.	No
NFR-002	Critical services shall be stateless where practical and horizontally scalable.	Must	Autoscaling test.	Partly
NFR-003	Subscriber-level ordering shall be preserved for fulfilment and repayment events where required.	Must	Ordering test.	No
NFR-004	The platform shall survive at least a 5x forecast burst without data loss, using queues and backpressure.	Must	Stress test.	Partly
NFR-005	Capacity dashboards shall show usage and headroom by telco and service.	Must	Observability test.	Yes

19. Security, Privacy and Compliance

- Zero-trust service authentication; mTLS for telco and internal high-trust integrations where appropriate.
- Encryption at rest and in transit; tenant-specific keys where required.
- Secrets managed through a vault/KMS, rotated without code change.
- Strong RBAC/ABAC using role, telco scope, product scope and action.
- MFA for privileged users; just-in-time and break-glass access controls.
- Tamper-evident audit logs and security monitoring.
- Tokenisation or hashing of MSISDN in analytical environments where feasible.
- Data minimisation, retention, deletion and legal-hold controls.
- Consent/disclosure records where required.
- Secure SDLC, dependency scanning, SAST, DAST, penetration testing and threat modelling.
- Fraud monitoring for credential abuse, replay, enumeration and automated request attacks.

ID	Requirement	Priority	Acceptance Evidence	Configurable?
SEC-001	Cross-tenant data access shall be prevented at API, service and database layers.	Must	Penetration and RLS tests.	No
SEC-002	Privileged configuration changes shall be signed, approved and audited.	Must	Audit test.	No
SEC-003	Logs shall not contain raw secrets, full NIN or unmasked sensitive fields.	Must	Log scan.	No
SEC-004	The platform shall support configurable retention by data class and tenant within legal constraints.	Must	Retention test.	Yes
SEC-005	Security incidents shall be isolatable by telco without shutting the entire platform where safe.	Should	Incident drill.	Partly

20. Observability and Operational Resilience

- Distributed tracing using correlation IDs spanning telco gateway, decision, fulfilment, ledger and notification.
- Metrics by telco, product, channel, adapter and status.
- Structured logs with masking and traceability.
- Business controls: approvals, fulfilments, recoveries, exposure, settlement variance and reconciliation ageing.
- Technical controls: latency, error rate, queue lag, retry rate, circuit-breaker state, saturation and database health.
- Synthetic probes for telco endpoints and internal critical paths.
- Runbooks for telco outage, batch failure, unknown fulfilment, ledger imbalance, queue backlog, data corruption and cyber incident.
- Regular disaster-recovery and reconciliation-rebuild exercises.

21. Edge Cases and Required Behaviour

Scenario	Trigger	Required Behaviour
Duplicate advance request	Same request ID replayed	Return original result; do not create second reservation/advance.
Concurrent requests	Two channels request simultaneously	Serialize or atomically reserve exposure; total cannot exceed limit.
Telco timeout after credit	Platform does not receive response	Mark FULFILMENT_UNKNOWN; query status/reconcile before retry.
Platform crash after telco success	Credit succeeded before local activation	Recover via outbox/inbox/reconciliation; activate exactly once.
Platform approves but telco rejects	No airtime/data delivered	Mark failed, reverse reservation and provisional entries.
Partial fulfilment	Telco delivers different amount	Quarantine; either recognise actual amount under approved rule or reverse/correct.
Duplicate recovery event	Telco replays recharge deduction	Idempotency prevents double repayment.
Out-of-order recovery/reversal	Reversal arrives first	Park until original arrives or resolve through exception process.
Recovery exceeds outstanding	Overpayment	Cap allocation; create unapplied item/refund/credit according to policy.
Recharge reversal after repayment	Previously recovered value reversed	Reverse allocation and reopen exposure.
SIM swap after score	Risk changes before borrow	Real-time overlay blocks/reduces offer for configured cooling period.
NIN flag changes to unverified	Eligibility lost	Block new advances; existing obligations remain recoverable.
Number porting	MSISDN moves operator	Close old telco service account, preserve history, create new telco account; no cross-tenant assumption.
MSISDN recycled	Number assigned to new person	Use telco effective dates/identity confidence; do not expose predecessor history.
One-off large recharge	Attempt to game limit	Spike discounted; max upward movement one configured tier.
Stale score	Batch delayed	Apply score-age policy: allow conservative offer, re-score or decline safely.
Missing feed fields	Partial telco data	Use approved fallback only; otherwise quarantine and protect prior valid data.
Corrupt batch file	Checksum/schema mismatch	Reject whole file or safe partition; alert; retain last valid snapshot.
Telco sends wrong tenant ID	Credentials and payload disagree	Reject and raise security alert.
Product disabled mid-request	Kill switch activated	Requests not yet fulfilled fail safely; active obligations unchanged.
Tier configuration gap	No matching tier	Fail configuration validation before activation; never guess.
Fee change during accepted offer	Offer accepted under old terms	Honour accepted offer version until expiry.
Funder limit exhausted	Portfolio capacity unavailable	Decline new originations or route to approved alternative funding pool.
Telco outage	Fulfilment unavailable	Open circuit breaker; suspend offers or queue according to configured safe mode.
Queue backlog	Events delayed	Scale workers, apply backpressure, preserve ordering and monitor SLA.
Database failover	Primary unavailable	Fail over without duplicate posting; validate ledger sequence and replay outbox.
Ledger imbalance	Posting template defect	Stop affected product/telco posting, quarantine transactions and alert finance.
Settlement dispute	Telco rejects statement item	Create case, freeze disputed amount only, preserve undisputed settlement.
Manual adjustment abuse	Agent attempts unauthorised change	Maker-checker, limits, reason/evidence, no ledger edits.
Customer dispute after repayment	Customer claims wrongful deduction	Provide trace, telco evidence and controlled

Scenario	Trigger	Required Behaviour
		refund/reversal workflow.
Time-zone/day boundary	Telcos use different zones	Store UTC plus telco-local business date; settlement based on configured calendar.
Currency mismatch	Wrong-currency event	Reject/quarantine; never auto-convert unless an approved FX process exists.
Schema version change	Telco sends new payload	Version negotiation; unsupported versions rejected with clear error.
Certificate expiry	Telco credential expires	Advance alerting and rotation; fail securely.
Notification failure	SMS/app message unavailable	Financial transaction remains valid; retry notification separately.
Analytics lag	Warehouse delayed	Operational decisions continue from operational store; dashboards show freshness.
Write-off then recovery	Late recharge received	Post post-write-off recovery; do not erase write-off history.
Death/inactive account/legal hold	Special treatment required	Apply approved status workflow and recovery restrictions.
Mass configuration error	Risk rule would over-approve	Simulation, change approval, canary rollout, automatic guardrails and rollback.
Cross-telco coincidence	Same MSISDN appears in historical data	Treat telco_id + effective period as distinct subscriber accounts.

22. Core Data Model

Entity	Key Attributes
TelcoTenant	telco_id, country, currency, time zone, status, deployment mode.
SubscriberAccount	subscriber_account_id, telco_id, normalised MSISDN token, effective dates, status.
SubscriberFeatureSnapshot	snapshot_id, source period, feature values, quality, schema version.
ScoreDecision	score_id, model/rule/config versions, contributions, reason codes, decision time.
CreditTier	tenant/product tier number, amount, movement rules, effective dates.
Offer	offer_id, subscriber, product, gross/net/fee/repayment, expiry, terms version.
Advance	advance_id, request/idempotency keys, product, amount, state, fulfilment reference.
ExposureReservation	reservation_id, amount, expiry, status.
LedgerEntry	entry_id, event, debit/credit account, amount, currency, references.
RecoveryEvent	recovery_id, telco event reference, source recharge, amount, status.
RepaymentAllocation	allocation_id, recovery_id, advance_id, component, amount.
ReconciliationItem	source, expected, actual, variance, match status and case.
SettlementPeriod	tenant/product/funder dates, status, totals and statement.
ConfigurationVersion	domain, scope, values, lifecycle, approvals and hash.
AuditEvent	actor, action, object, before/after hashes, time, reason, source IP/device.

23. Testing and Quality Assurance

- Unit and property-based testing for scoring, tier movement, fees, allocation and ledger invariants.
- Contract tests for each telco adapter and schema version.
- End-to-end tests covering USSD/app request through fulfilment, repayment, reconciliation and settlement.
- Load, soak, burst and capacity tests using realistic telco distributions.
- Chaos tests for endpoint timeout, duplicate events, queue failure, database failover and partial region loss.
- Security, penetration, tenant-isolation and privilege-abuse tests.
- Model validation, back-testing, bias/fairness review where applicable, drift and stability tests.
- Financial reconciliation tests proving transaction-level totals.
- Disaster-recovery, restore and historical replay tests.
- UAT by risk, operations, finance, support, security and each telco.

ID	Requirement	Priority	Acceptance Evidence	Configurable?
TST-001	No release shall proceed with unresolved critical ledger, tenant-isolation or duplicate-processing defects.	Must	Release gate.	No
TST-002	Each telco adapter shall have automated certification tests before production activation.	Must	Certification report.	No
TST-003	Configuration changes shall be testable in simulation against historical or synthetic data.	Must	Simulation evidence.	Yes
TST-004	Performance tests shall use agreed peak TPS and 5x burst scenarios.	Must	Performance report.	Yes

24. Telco Onboarding and Migration

12. Commercial and regulatory readiness confirmed.
13. Tenant created with country/currency/time zone and data-residency settings.
14. Data contract and canonical mappings agreed.
15. Connectivity, authentication and certificate exchange completed.
16. Historical subscriber features and incumbent loan data mapped and quality-assessed.
17. Adapter contract tests and security tests passed.
18. Products, tiers, fees, scoring, recovery and settlement configured in draft.
19. Historical back-testing and limit distribution approved by risk and telco.
20. Shadow scoring and reconciliation run without customer impact.
21. Pilot cohort enabled using feature flags and conservative caps.
22. Ramp-up by segment/region/percentage with live monitoring.
23. Full launch after operational, financial and risk acceptance.

Migration from an incumbent provider shall preserve external references, outstanding advances, repayment history, customer treatment and settlement positions. The platform shall support dual-running and cutover reconciliation.

25. Reporting and Analytics

- Daily operational report by telco/product/channel.
- Portfolio exposure, approvals, take-up, utilisation and available funding.
- Repayment curves, delinquency, roll rates, write-offs and recoveries.
- Vintage/cohort analysis and tier migration.
- Fee, revenue-share and profitability reporting.
- Reconciliation and settlement variance ageing.
- Data quality and feed freshness.
- Model performance, drift and reason-code distribution.
- Fraud alerts and suspicious recharge/borrow behaviour.
- Regulatory/consumer complaint and service-availability reports.

26. Regulatory and Policy Considerations

Final legal classification, licensing, customer-disclosure, fee, consent, data-processing, complaints, credit reporting and collection requirements shall be confirmed for each jurisdiction and product before launch. The system shall therefore support configurable disclosures, consent capture, fee caps, cooling-off or grace rules, complaint SLAs, data-retention and audit evidence. Legal requirements shall not be assumed from a prior airtime-advance model.

27. Deployment and Technical Architecture Guidance

- Containerised services orchestrated across multiple availability zones.
- API gateway/WAF, Telco Integration Gateway and independently deployable telco adapters.
- Separate services for subscriber profile, scoring, offer, advance, fulfilment, ledger, recovery, reconciliation, settlement, configuration, notifications and identity/access.
- Durable event bus with partitioning, schema registry, dead-letter queues and replay controls.
- Relational transactional store for advances/ledger; distributed cache for offers; object storage/data lake for raw feeds; warehouse/lakehouse for analytics.
- Outbox/inbox patterns for reliable state-event consistency.
- Immutable backups, point-in-time recovery and cross-region disaster recovery for critical data.
- Infrastructure as code and environment-specific configuration; no secrets in source control.

28. Release Acceptance Criteria

- At least one telco completes end-to-end certification and a second telco can be configured without core-code changes.
- All financial transactions reconcile from request through settlement.
- Duplicate, timeout-after-success and recovery-reversal scenarios pass.
- One-tier maximum movement and anti-gaming scenarios pass.
- Tenant isolation, RBAC and security tests pass.
- Capacity and latency targets are achieved at forecast peak and burst load.
- Disaster recovery and ledger reconstruction are demonstrated.
- Admin configuration lifecycle, approval, simulation and rollback are demonstrated.
- Operational teams have dashboards, alerts, runbooks and trained users.
- Risk, Finance, Security, Compliance, Product and Telco stakeholders sign off.

29. Delivery Workstreams

Workstream	Principal Deliverables
Platform Foundation	Tenant model, identity, configuration, audit, observability and deployment.
Telco Integration	Gateway, adapters, batch/stream/API contracts and certification harness.
Data and Scoring	Feature store, scoring pipeline, anti-gaming, tiering and explainability.
Credit Core	Offers, advance lifecycle, exposure, concurrency and fulfilment orchestration.
Ledger and Recovery	Sub-ledger, recovery allocation, reversals and write-offs.
Reconciliation and Settlement	Matching, exceptions, commissions, statements and finance exports.
Portals	Admin, operations, risk, finance, support and telco views.
Security and Compliance	IAM, encryption, privacy, regulatory controls and testing.
Operations Readiness	Runbooks, monitoring, DR, support model, SLAs and training.
Telco Launch	Data mapping, back-test, shadow run, pilot, ramp-up and cutover.

30. Decisions Required Before Detailed Design Freeze

Decision	Options/Questions
Legal lender/funder	Platform company, licensed lender, bank, telco or multi-party structure?
Fulfilment mode	Platform calls telco credit API or telco fulfils after platform approval?
Recovery mode	Telco autonomously deducts and reports, or platform instructs deduction?
Fee structure	Upfront deduction, added repayment amount, bundle-specific pricing, tax treatment?
Data frequency	Daily full feed, incremental intraday, streaming events and API fallback?
Infrastructure location	Cloud, telco-hosted, hybrid, country-specific or dedicated tenant?
Database isolation	Shared partitioned, schema-per-telco or dedicated for major telcos?
Initial products	Airtime only, airtime + data, or broader digital credit catalogue?
Target scale/TPS	Subscriber counts, peak USSD/app TPS, recharge event peaks and scoring windows?
Regulatory controls	Required licences, disclosures, consent, complaints and reporting?

Appendix A - Glossary

Term	Meaning
Advance	Credit value delivered now and recovered later.
MSISDN	Subscriber mobile number identifier.
Telco	Mobile network operator or MVNO connected to the platform.
Tenant	Logically isolated telco/operator configuration and data scope.
Offer Snapshot	Pre-calculated, time-bound set of amounts available to a subscriber.
Exposure	Outstanding unrecovered obligation.
Idempotency	Repeated processing produces only one financial effect.
NIN Verification Flag	Boolean/status confirmation without necessarily sharing the actual NIN.
Reconciliation	Comparison of records between systems to identify and resolve differences.
System of Record	Authoritative source for a defined data domain.
Tier	Configured credit-limit band used for progressive trust.
Outbox/Inbox	Reliable messaging patterns linking database state and event publication/consumption.

Appendix B - Configuration Guardrails

- No configuration may bypass ledger balancing, tenant isolation, idempotency or authentication.
- High-risk changes require dual approval and may require risk/finance/security approval based on domain.
- All values must have allowed ranges and semantic validation.
- Changes affecting approval rates or exposure require impact simulation and optional canary rollout.
- Rollback must restore the previous version without deleting historical decisions.
- Emergency suspension is immediate but reactivation requires normal approval unless an approved incident procedure applies.